

Ignacio Jolín Rodrigo

Mathematician

About Me

I am a Mathematician (BSc & MSc) and AI Researcher working at the intersection of geometric machine learning, topological methods, and modern language models. My mathematical background drives my focus on understanding the underlying structure, geometry, and invariant features of high-dimensional data, moving beyond black-box heuristics.

My research develops across two complementary pillars. First, in Topological Data Analysis (TDA), I specialize in persistent homology, simplicial and cubical complexes, and persistence landscapes to capture the underlying geometry of complex data. Second, I work on the empirical evaluation, alignment, and deployment of Large Language Models (LLMs) in high-stakes domains. Across both fields, my goal is to bridge rigorous mathematical theory and practical computation, building reliable systems grounded in structural guarantees rather than black-box heuristics.

Resume

Experience

Researcher In The Universidad Alcalá De Henares, Madrid
Researcher Of Generative AI Applied To Health
September 2025 - September 2026
- **API Integration:** Engineered scalable solutions by integrating third-party LLM APIs into production workflows to enhance system capabilities.
- **Infrastructure & Deployment:** Deployed and optimized open-source LLMs (e.g., Llama, Mistral) on private, on-premise servers, ensuring data sovereignty and reduced latency.
- **NLP Evaluation Frameworks:** Designed and implemented quantitative metrics and benchmarking protocols to rigorously assess the quality, accuracy, and coherence of LLM-generated summaries.
- **Scientific Dissemination:** Authored and structured high-impact scientific publications utilizing LaTeX for precise document formatting and technical composition.
- **Cross-Disciplinary Collaboration:** Partnered with clinical stakeholders in a hospital setting to align technical AI solutions with healthcare requirements and patient data privacy standards.
- **Large Language Model (LLM) Optimization:** Executed fine-tuning strategies on foundational models to tailor performance for specific domain tasks.

SW And Technology Department At Renault I+D+I, Valladolid
Data Science
September 2024 - June 2025
- **Linux:** Worked with Linux on Raspberry Pi, implementing communication protocols.
- **Data Science:** Developed image classification models and built a predictive model for gas consumption forecasting.
- **Documentation:** Created technical documentation using Jupyter Notebooks and Markdown.

Education

Universitat Politècnica De València, Spain
Master's Degree In Mathematical Research
2023 - 2024
My master's program equipped me with a dual perspective: the rigorous methodology of academic research and the pragmatic mindset of industry applications. I developed the ability to navigate between theoretical frameworks and real-world problem-solving, mastering how to adapt abstract concepts into practical solutions. This experience improved my capacity to work effectively in both environments, whether conducting structured research or delivering actionable results under technical and time constraints, while maintaining the precision of mathematical thinking at the core of my approach.

Universidad De Valladolid, Spain
Bachelor's Degree In Mathematics

2019 - 2023
I mastered the foundational principles of diverse mathematical fields, from pure to applied disciplines. Beyond technical knowledge, this rigorous training honed my ability to think in abstract terms, construct logical frameworks, and solve complex problems with structured creativity. These core skills now serve as my toolkit for tackling challenges in AI research and data science, where deep mathematical reasoning meets real-world impact.

Erasmus Exchange Program At The Universidade De Coimbra, Portugal

Bachelor's Degree In Mathematics

2021 - 2022
This marked my first experience living far from home, where I embraced the challenge of adapting to a new cultural environment. I learned to collaborate with people from diverse backgrounds, explored fascinating academic topics beyond my usual scope, and grew by stepping far outside my comfort zone.

Languages

English
Certified with B2 First – Cambridge

Spanish
Native

Chinese
Certified with HSK2

Portuguese
Elementary

Hard Skills

Coding
RStudio, MATLAB, Mathrmatica, C, Bash, Python (Threading, Multiprocessing, Pillow, Keras, TensorFlow, PyTorch, Numpy)

Mathematics
Topology, Optimization, Functional Analysis, Algebra, Mathematical Modelling, Statistics, Cryptography

Machine Learning
With a strong theoretical background: Artificial neural networks, DImension reduction, Random forest, Convolutional neural networks, Support vector machine, Large Languages Models

OS
Linux

Version Control
Git

Technical Redaction
LaTeX, Markdown, Jupyter

Databases
SQL

Soft Skills

Abstract Problem Resolution

Effective Communication

Quick Learning Ability

Academic Research

Team Work

Publications

2026

Evaluating LLMs For HIV Epidemiological Surveillance In Spanish: Clinical Summarization, Risk Factor Extraction, And Inference Stability Across Models.
Preprint under review, Research Square, DOI:10.21203/rs.3.rs-10166213/v1 (accessed 2026-09-04), Universidad de Alcalá de Henares, Spain.
Ignacio Jolín, Ana Carrero-Fernández, Juan Cuadros-González and Maria D.R-Moreno

2025

Time Series Analysis With Persistence Landscapes: A Novel Model For Alzheimer Detection
Preprint, Accessed: 2024-07-15. DOI: 10.36227/techrxiv.174494666.69176037/v1, Valladolid, Spain
Ignacio Jolín

2024

Master's Thesis: Clasificación De Series Temporales Empleando Análisis Topológico De Datos
Valencia, Spain
Ignacio Jolín, Antonio Falcó

Introduction To Knot Theory
XI Congreso del Máster en Investigación Matemática y Doctorado en Matemáticas, Universitat de València, Spain
David Campos-Abad, **Ignacio Jolin-Rodrigo**, Clara Soler-Signes

2023

Bachelor's Thesis: Lema De Sperner Y Topología Simplicial
Valladolid, Spain
Ignacio Jolín-Rodrigo, Jesús Dominguez

Projects

Multiparameter Persistence In Medical Imaging
Implementation of 2D cubical bifiltrations via distance transforms (SEDT), admissible 1D slicing, and Bubenik landscapes with GUDHI on RetinaMNIST.
`Git: Multiparameter_persistence`

GPT From Scratch
In this repository, I explain most of the mathematical concepts from a GPT model, explaining how the transformers and other concepts work
`Git: GPT_from_scratch`

Reinforcement Learning Agent
A DQN agent that navigates a square toward a target circle on the screen avoiding obstacles
`Git: Reinforcement-Learning`

Dimension Reduction Theory
Jupyter file with the theory of the main dimension reduction algorithms
`Git: Dimension-Reduction-Theory`

Data Science Tips
Jupyter file with some tips for data science and practical examples
`Git: Data-Science-Tips`

Chinese App
Android app that helps you learn Chinese
`Git: App_chino`